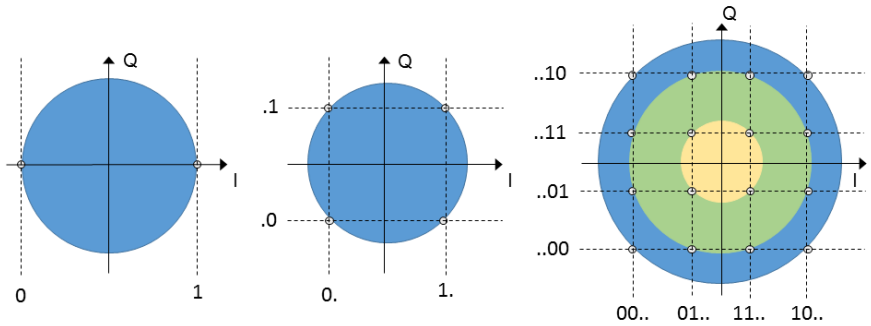
Solution 1

Discussion: 06.05.2026

Solution 1.1: Signals, Modulation, and Bandwidth

a.) *Modulate the ASCII character sequence **MIT26** with the following modulation schemas:*



For your solution, assume that each signal has a duration of one oscillation. For representation of the ASCII characters, refer e.g. to wikipedia.

ASCII: MIT26 = 77 73 84 50 54 = 01001101 01001001 01010100 00110010 00110110

- For a), encode each bit itself
- For b), encode 01 00 11 01 01 00 10 01 01 01 01 00 00 11 00 10 00 11 01 10
- For c), encode 0100 1101 0100 1001 0101 0100 0011 0010 0011 0110

Solution see video recording or hand-drawn solution.

b.) *You are designing a wireless communication system that needs to operate with 9.6 kBit/s. You have chosen 16-QAM for modulation. Which bandwidth do you have to assign to your system?*

Given: R = 9.6 kBit/s; number of signal levels n = 16

Nyquist: R [bit/s] $\leq 2 \cdot B$ [Hz] $\cdot$ ld(n) [bit]

Solve for bandwidth B: $B \geq R/2\text{ld}(n) = 9.6/2 \cdot \text{ld}(16) = 1.2$ kHz

c.) *Given is a channel with a bandwidth of 3 MHz. We want to achieve a data rate of 20 MBit/s. What minimum SNR do you need to achieve this data rate? Give its value it dB.*

You have given: R = 20 MBit/s; B = 3 MHz.

Shannon: R [bit/s] $\leq B$ [Hz] $\cdot$ ld(1 + S/N) [bit]

Solve for SNR: $S/N \geq 2^{R/B}-1 = 2^{20/3}-1 = 100.59 \approx 100$

In dezibel: $SNR_{dB} = 10 \cdot \log_{10}(100) = 20$ dB

When having to talk about antenna amplification, attenuation, or noise, you always have given the SNR in dB, thus it is very helpful to have an understanding of its meaning.

d.) *You are able to guarantee a SNR of 20dB for the channel above. Which modulation you can apply on the given channel? Can you achieve the target data rate of 20 Mbit/s? If not, what changes in your system you might do to achieve the target data rate?*

The question is connected to part c.): we have given a bandwidth of 3 MHz. With this bandwidth and the SNR, using Shannon theorem we first can calculate if we can achieve 20 Mbit/s. This we already have done in part c.): yes, we can achieve 20 Mbit/s, but not more!

Now use Nyquist for getting the number of signal levels you need to come to 20 Mbit/s. For doing so, solve the equation for n: R [bit/s] $\leq 2 \cdot B$ [Hz] $\cdot$ ld(n) [bit].

$n \geq 2^{R/2B} = 10.08$

Ah – you need at least 10.08 levels, then. That's kind of not really possible. But, using 11 levels already results in more than 20 Mbit/s, and due to Shannon this is not possible (the error rate would be to high for communication). Thus, 16 levels (by using 16-QAM as next specified schema) are impossible as well. You have to restrict to a code with 8 states (e.g. 8-PSK) and have to see how to adopt the bandwidth (the signal rate) to come to your target data rate:

Solve R [bit/s] $\leq 2 \cdot B$ [Hz] $\cdot$ ld(n) [bit] for the bandwidth:

$B \geq R/2 \cdot \text{ld}(n) = 20/6 = 3{,}33$ MHz.

Or instead, try to improve the SNR so that you can make use of 16-QAM, e.g. by using special antennas with a gain. If you want to stay with 16-QAM, you are achiving a data rate of R [bit/s] $\leq 2 \cdot B$ [Hz] $\cdot$ ld(n) [bit] = 24 Mbit/s. You now can calculate the SNR as above as $S/N \geq 2^{R/B}-1 = 255$(= 24 dB). Thus, you need an antenna gain of about 2.5.

Note: this calculation is not that realistic. Please only keep it as a simplified connection between Shannon- and Nyquist Theorem. In practice, you always have to use Forward Error Correction because errors are present all the time. Thus, you can never achieve the full data rate but you have to spend some fraction for FEC. On the other hand, when using FEC, in the above scenario you might use 16-QAM together with FEC to approximate the 20 Mbit/s.

Solution1.2: Signal Propagation

Assume a connection between a mobile device and a Wi-Fi access point in an 802.11n network on channel 1. The transmit signal power is 100 mW and the receiver requires a signal power of at least 10^{-8} mW. The gains G_i of the antennas are 2.15 dbi. (HINT: find out the center frequency of channel 1 in 802.11g. And: dbi is a **db**-measure referring to an **i**sotropic radiator as a reference – while in your calculations you would have to use the direct relationship, not the db-representation.)

a.) Compute the maximum possible communication distance between the two devices in a free space setup. How does the distance influence the signal that is perceived at the receiver?

b.) By which ratio does the maximum propagation distance decrease assuming a city scenario with $\alpha = 3.5$? Which effects are influencing the signal strength?

c.) *How does changing to channel 140 in the 5 GHz band influence the signal propagation distance?*

a.) to c.) Use formula of slide I-47 and solve for *d*. (As done on handwritten solution.) You have given all parameters – just don't forget to transform the antenna gain in absolute values (it's given in dBi) and have a look about the center frequency of Wi-Fi channel 1 to calculate the frequency of it. See excel table for calculating with arbitrary values.

In the solution to a.) the distance is far higher than you experience in practice – because still in a free space setup you don't have a power of 2 with which the signal strength decreases with distance. Due to noise and attenuation by the atmosphere as you always have a power of more than 2. In typical setups you additionally have multipath effects. And the simple formula anyway only holds in idealized environments, neglecting loss of powers by imperfect hardware (antenna, cables, connectors) and other factors.

Thus: use the outcomes only as approximation with large variances.

d.) *Why are waves with very low frequency and waves with very high frequencies (higher than some hundred GHz) not used for data transmission in computer networks?*

There are several reasons why low frequencies are not used for radio data transmission:

- Low frequencies give us only low data rates, because the bandwidth is smaller than for higher frequencies: the carrier frequency always has to be higher than the frequency of the carried signal – in more detail: than the frequency of the highest needed component in your restricted baseband signal. Thus, the bandwidth always is lower than the carrier frequency – and thus, using very low frequencies as carrier drastically restricts the bandwidth and as a consequence, the data rate.
- Furthermore: for generating and receiving low frequencies we need large antennas (remember the relation between wavelength and antenna size) – mobility is very restricted with antenna sizes for very low frequencies....
- The attenuation and shading of low frequencies is not very high – so (see the AM radio broadcasting) we have interference ranges of hundreds of kilometres, which is not useful for setting up a cell-based network.

Waves with THz-frequencies are in or near the range of infrared and visible light:

- These signals are shaded by nearly any material, thus transmission is only possible line-of-sight. And only within some meters, afterwards the signal is completely attenuated. The range is by far too small even for local networks.
- Furthermore, using waves of/near infrared/visible light for transmission is not very robust: even a simple lamp would produce lots of interferences.
- And: even using the THz range below infrared is not prominent, because it is technically not simple to produce waves of that range.

Solution 2

Discussion: 18.05.2026

Solution 2.1: Multiplexing

For multiplexing, SDM, TDM, FDM, and CDM were presented in the lecture.

a.) *CDM systems are often called "much more flexible" than TDM or FDM systems. Reason about that: what limits the number of simultaneous users in a TDM/FDM system compared to a CDM system? How does an additional user influence the other users in the cell in FDM, TDM, and CDM systems?*

In TDM/FDM systems, we have time slots on certain carrier frequencies which can be assigned to single users. When all slots / all carriers are assigned, no new users can be accepted – we have a fixed maximum number of users. Dynamics like re-definition of slot size or carrier bandwidth are not specified for these methods because it is physical changes. Within CDM systems, we have dynamics integrated, giving a soft limit of the number of users: just choose longer codes for a higher number of users – by a software configuration, which is by far more simple.

But: if with each new user longer codes are needed, the data rate per connection goes down drastically. Also, if only quasi-orthogonal codes are used the noise level for all others is increased with each new user, in extreme case resulting in cell breathing (see UMTS chapter later on for details) which limits the number of simultaneous users. In contrast, in TDM/FDM systems users cannot influence each other.

In practice, also a certain data rate is mandatory – thus in CDM systems flexibility is additionally reduced by defining a maximum possible code length.

b.) *In GSM a combination of TDMA and FDMA is used. Thus we as well have to keep guard spaces as to provide synchronization – we combine the disadvantages of both schemas! Why is such a combination nevertheless done instead of using FDMA or TDMA only?*

In systems like GSM we have to have a certain data rate (voice transmission) for each station. We could make use of FDMA to create channels of minimum necessary bandwidth to give this data rate. But, the narrower a channel is, the more susceptible to narrowband interference it is. Thus we want to have broader channels. Other extreme case: use TDMA on a single channel only. But, to keep the same data rate as before over this broad, slotted channel would give need for having a high signal rate – which increases the risk of inter-symbol interference.

Thus: use FDMA and TDMA in combination to give relatively broad channels but nevertheless not much ISI problems. (Also there are some other reasons, e.g.: using only FDMA would give the demand for two antennas when establishing a duplex communication. Using time slots on one carrier to separate uplink and downlink, you only need a single antenna which switches between transmitting and receiving.)

c.) *What are the main benefits of a spread spectrum system?*

Spread spectrum technology offers robustness against narrowband interference, and against multipath propagation.

Maybe – but not important for us – you might mention protection against tapping or (when using it for medium access coordination), more dynamics in the maximum number of simultaneous users.

d.) *You want to set up a CDMA-based system for two users. The symbol rate is given with r_s = 1 MSymbols/s. PSK is used for modulation. Which orthogonal codes you could choose to achieve a data rate of r_1 = 500 kBit/s for user 1 and a data rate of r_2 = 125 kBit/s for user 2? Is it possible to later on add a third user who needs a data rate of 250 kBit/s?*

User 1 has to send one bit each two chips. Thus you could choose (1, 1) as a code for this station. User two only has to send one bit all eight chips, thus we can choose any chipping sequence of 8 bit which is orthogonal to the first one in each sub-sequence of 2 bits, e.g. (1, -1, 1, -1, 1, -1, 1, -1).

We can add a third user later on if we find a code of length 4 which is orthogonal to both above codes, e.g. (-1, 1, 1, -1).

e.) *Given are the following chip sequences:*

A: (-1 -1 -1 +1 +1 -1 +1 +1)

B: (-1 -1 +1 -1 +1 +1 +1 -1)

C: (-1 +1 -1 +1 +1 +1 -1 -1)

D: (-1 +1 -1 -1 -1 -1 +1 -1)

The stations want to transmit the following bits: A: 1, B: 0, C: 0, D: -. What is the transmitted sequence?

A transmits its inverted chip sequence, B and C transmit the chip sequence, D transmits nothing:

A	+1	+1	+1	-1	-1	+1	-1	-1
B	-1	-1	+1	-1	+1	+1	+1	-1
C	-1	+1	-1	+1	+1	+1	-1	-1
D	0	0	0	0	0	0	0	0
	-1	**+1**	**+1**	**-1**	**+1**	**+3**	**-1**	**-3**

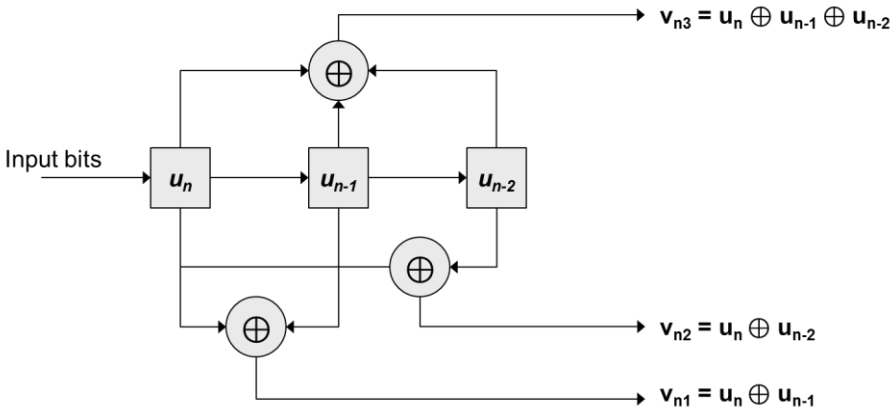
Solution 2.2: Forward Error Correction with Convolutional Codes

Given is a (3,1,3) convolutional code with

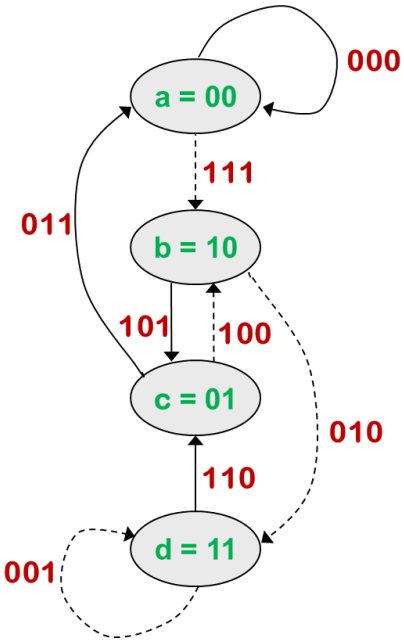
$v_{n1} = u_n \; XOR \; u_{n-1}$, $v_{n2} = u_n \; XOR \; u_{n-2}$, $v_{n3} = u_n \; XOR \; u_{n-1} \; XOR \; u_{n-2}$.

a.) Give the shift register diagram and the state diagram for this code.

Shift register:



State diagram:

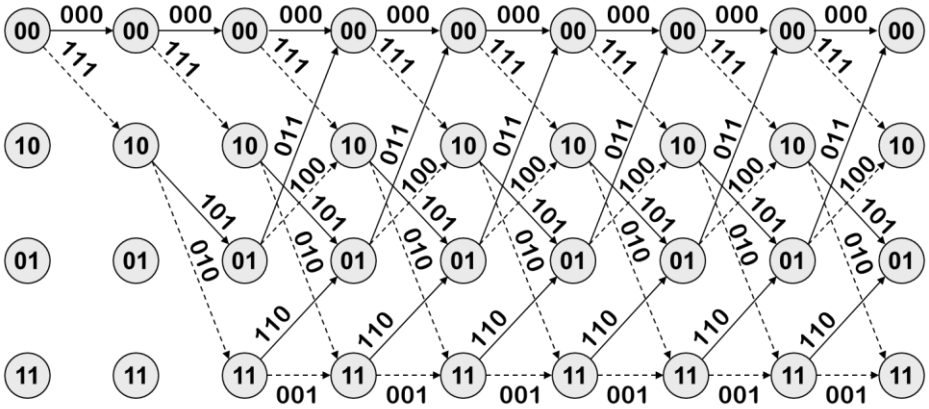


b.) *Assume that the bit sequence 1 1 1 0 1 is to be encoded. Give the output of the encoder.*

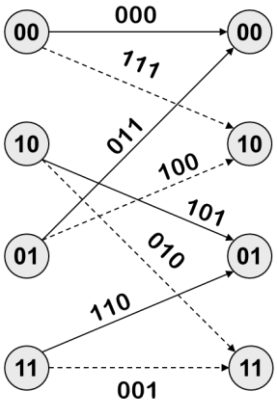
Output: 111 010 001 110 100

c.) *Give the Trellis diagram for the code. What is the free distance of the code?*

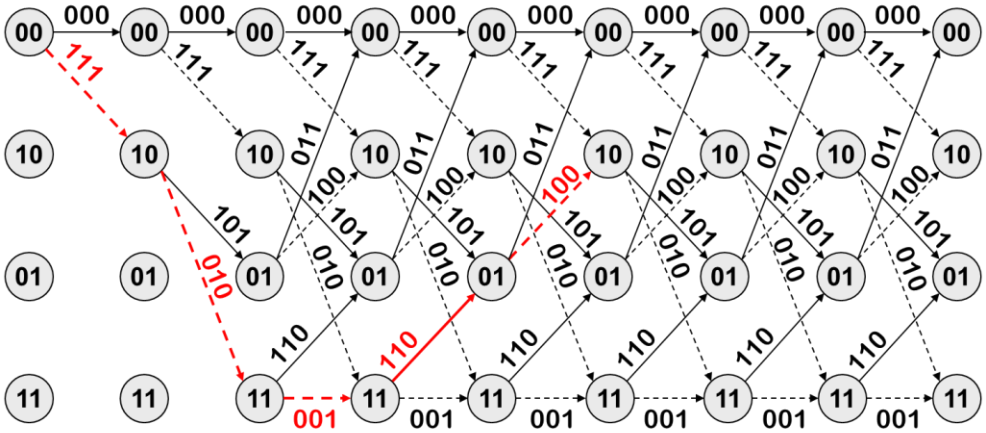
Trellis diagram:



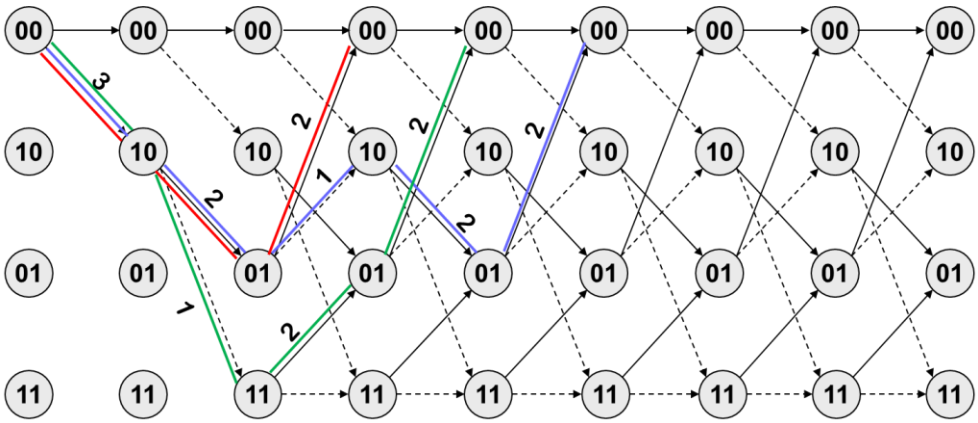
Kind of complex. Shorter written (then just combine):



Hey, look at part b), you can also read out the answer here:



Free distance: find cycle (from (0,0) to(0,0)) with smallest Hamming distance to the zero path:

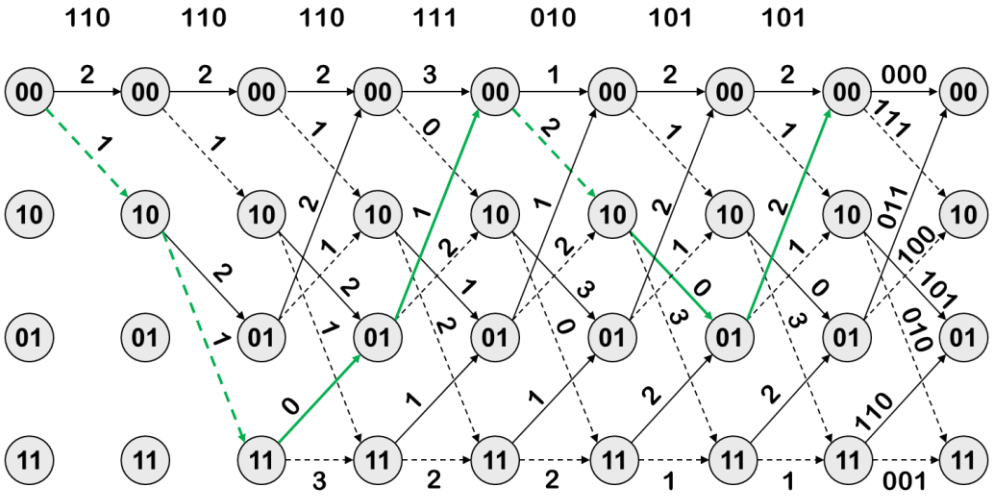


The red path has smallest Hamming distance (that is, the smallest number of 1s). Sum up: the free distance is 7.

The blue path is kind of redundant. You would also have to check it to cover all possible paths (thus, take each transition from each state at least once), but we end up in a formerly visited state here. Thus, we only can end up with a larger number of 1s while searching for the minimal number. (Same holds for the second transition from state (1,1): we don't leave this state so we don't come closer to the (0,0) state but just increase the number of 1s, thus we can ignore it.

d.) *You are receiving the sequence 1 1 0 1 1 0 1 1 0 1 1 1 0 1 0 1 0 1 1 0 1. Find the most likely bit sequence that was transmitted using the Viterbi algorithm in terminated mode.*

Terminated mode: path has to end in (0,0). Calculate shortest path after seven transitions. You have to look to all possible paths to this state; here only the shortest one is shown:



The output sequence is corrected to 111 010 110 011 111 101 011 which corresponds to the input 1 1 0 0 1 0 0.

Doing this task, maybe you experience how hard it is to do the corrections efficiently…

Solution 3

Discussion: 22.05.2026

Solution 3.1: CSMA and MACA

Six stations are grouped around a mountain in a chain topology. Each station is able to hear only the two neighboured stations (station 1 and 6 can also hear each other). Stations perform CSMA, but without collision handling. Consider the following scenarios:

(i) Station 2 currently is sending to station 1. Station 3 wants to address station 4. Can station 3 start transmitting, and should/could it be allowed to do so?

(ii) Station 3 currently is sending to station 2. Station 5 wants to address data to station 4. Can station 5 start transmitting, and should/could it be allowed to do so?

(iii) Stations 1 and 2 are currently sending. Which other stations belief that they can start transmitting, and which ones should/could actually be allowed to do so?

(iv) Stations 1 and 4 are currently sending. Which other stations belief that they can start transmitting, and which ones should/could actually be allowed to do so?

(i) CSMA: Station 3 cannot start transmitting because it detects the transmission of station 2 when doing carrier sense. But, it could be allowed to do so because it doesn't interfere with 2's signals at 1, and 2's signals do not interfere with 3's signals at 4. We have a classical exposed station problem.
If MACA is used, we might solve this: station 3 has heard an RTS of station 2 before, but no CTS of station 1. We know that we are an exposed station and thus could start sending. But nevertheless it makes no sense sending an RTS: station 3 cannot receive the corresponding CTS of station 4 because it interferes with the transmission of station 2.
Thus: even if we are able to reason about our status as exposed station, we cannot deal with the problem.

(ii) CSMA: Station 5 does not detect the transmission of station 3 and thus it starts transmitting. But at station 4 a collision with the transmission of station 3 occurs, station 4 does not receive anything. Thus, it does not do any harm if station 5 starts transmitting, but it makes no sense, either. It would be best (scare resources, save energy) if station 5 would not start transmitting.
If MACA is used, station 5 would first send an RTS, but station 4 cannot receive it and thus does not answer. Station 5 would get no CTS and thus not start transmitting.

(iii) CSMA: Station 3 and 6 either are receivers, or they are sensing a communication between stations 1 and 2 (both stations started transmitting simultaneously). Stations 3 and 6 anyway cannot start another transmission. Only station 4 and 5 sense the channel idle and might start transmitting. But, their transmissions collides at station 3/6. Thus, at most stations 4 and 5 can start communicating with each other, and they only should do so if 3 and/or 6 are currently not receivers.
Using MACA, we know much better about what's going on. When stations 1 and 2 transmit simultaneously, they cannot communicate with each other because there must

have been a RTS/CTS handshake before. So, both initiated a communication to 3/6 simultaneously before. Thus stations 4 and 5 have received a CTS before, and both know that they are hidden stations and block.

(iv) CSMA: All (not-transmitting) stations are receiving some transmission (either they are receivers, or they sense an ongoing transmission). None starts transmitting. Starting anyway does not make any sense because no station would be able to receive another transmission without interference. Thus all stations should stay quiet to save energy.

With MACA that is possible – the two stations which are neither sender nor receiver know if they are exposed stations or not (depending if they received a CTS before the transmission or not). But, if they would try to start an own transmission, they would not get back any CTS because of interference.

Concluding: MACA solves the hidden station problem. It also allows stations to detect if they are exposed stations, but anyway does not solve the exposed station problem.

Solution 3.2: Hidden and Exposed Stations

a.) *Again consider the problem of hidden and exposed stations. Do you have hidden or exposed stations if Aloha, slotted Aloha, reservation Aloha, or MACA is used?*

- In Aloha, hidden stations are no special problem – all other stations are a problem. Stations start sending their data on demand, having no idea about the medium's or the receiver's status. They only learn about the collision later on by a missing ACK, when it is too late. So you could say, there are no hidden stations because anyway all stations just transmit when data is given, or you oculd say in principle hidden stations exist but they are not causing special problems. Exposed stations don't exist – a potential sender does not listen to the medium before sending, thus it cannot be exposed. Only when combining it with CSMA, we would have exposed stations.
- For Slotted Aloha, it is the same as for traditional Aloha. We just introduce slots to reduce the effects of collisions, but the transmission principle is the same.
- For Reservation-Aloha, both problems don't exist. Because we work with reservations, all other stations know about our as well as their own sending times. Only in the reservation phase, we can have problems with placing reservations, but because we will recognize if someone else tried to do the same reservation as we, the other stations are not hidden to us. (The hidden station problem by definition only might occur if data is sent. Reservations are control, not data, thus a collision in the reservation phase does not count for the hidden station problem.) The exposed station problem does not occur because in a reservation system typically all stations receive all transmissions, so it cannot be possible that two stations transmit simultaneously.
- MACA was designed to avoid both, hidden and exposed stations by "forwarding" carrier information with RTS and CTS control messages. See former question: hidden station problem solved, exposed station problem not.

b.) *How can MACA still fail in presence of hidden stations?*

When stations move, new connections (in the meaning of "in reception range of another station") can arise all the time and old ones can break. A mobile station might come into an existing network immediately after the exchange of the RTS/CTS messages, and if a current sender is hidden to this 'new' station, it can cause a collision, not knowing about the announced communication.

Furthermore, due to increase or decrease of signal strengths based on propagation characteristics, a connection between two stations in both directions can have completely different signal strengths. It is possible that one station does not receive the CTS of another one, but still can interfere the other station with its transmissions.

And still in case of ideal circumstances, stations can transmit with different power, having different communication ranges.

On such reasons the RTS/CTS mechanism is only suitable for more or less deterministic propagation characteristics and static devices. Ad-hoc networks can't make use of it.

Exercise 3.3: WLAN and OFDM

a.) *In 802.11a, OFDM with 52 subcarriers is used. 4 of these subcarriers are used as pilots. What is the purpose of such a "pilot" and why are there four (and not only a single one)?*

The four carriers are used as phase reference. Well-defined signals are transmitted here, to allow the receiver to synchronize signal detection and to estimate propagation characteristics and adapt the received (distorted) signals on the other carriers correspondingly.

We cannot live with only a single one because there always is the risk of narrowband interference which could destroy a carrier's information. Also, the distortion can vary over the whole bandwidth so that by having a look to all of them, the characteristics of the channels in between could be better estimated.

b.) *48 OFDM subcarriers are used in 802.11a, and modulation up to 64-QAM is applied. One OFDM signal has a length of 3.2 µs, and between two OFDM signals there is left a guard time of 0.8 µs (for avoiding inter-symbol interference). Which maximum data rate could be achieved? Why is it higher than the specified data rate of 54 MBit/s?*

With 64-QAM: coding of $N = 6$ bits in one signal.

Total time interval for one signal: $T_S = 4$ µs.

Parallel transmission on $S = 48$ subcarriers.

Thus: $r = \frac{N*S}{T_S} = \frac{6*48\ bit}{4\mu s} = 72MBit/s$

It is higher than 54 MBit/s because forward error coding is used. To achieve 54 Mbit/s, the SNR has to be high enough to modulate with 64-QAM and to apply error coding with code rate ¾.

Assignment 4

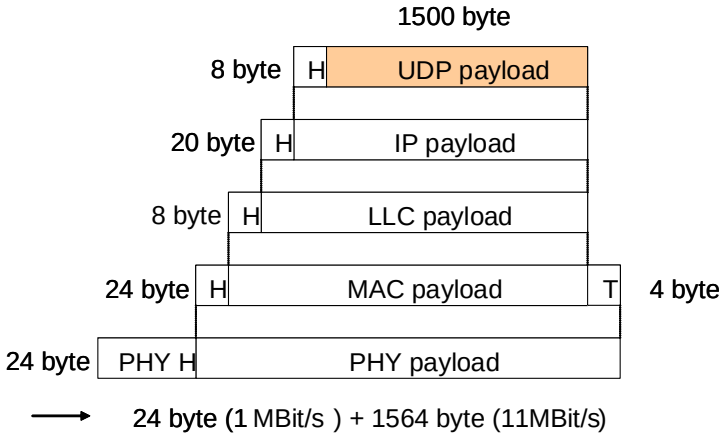
Discussion: 12.06.2026

Exercise 4.1: WLAN performance

The efficiency of a network heavily depends on its MAC. WLAN is one of the most prominent wireless networks today, but how efficient is medium access? Exemplarily, have a closer look to the standard 802.11b.

Let the traffic in an 802.11b network consist of UDP packets each containing 1500 Bytes of payload. Assume long frame format (see slide III-77). The maximum bit rate of 11 MBit/s can be maintained between all wireless stations and the AP. Estimate the maximum throughput (i.e. payload data rate) a station can achieve under these conditions. (Note: there are several aspects in the calculation which could be considered differently or depend on configuration of the network. Just give an approximate solution.)

Sent is the following data structure:



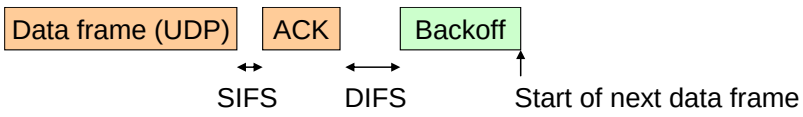
Note: there is also some additional LLC/SNAP information included (e.g. necessary for forwarding frames – Ethernet frames characterize their payload, many other frames do not…), and the MAC header only has 24 byte because the 4th address field is not needed. Thus:

- PHY frame duration: 24·8 µs (PHY H) + 1564·8/11 µs (PHY payload) = 1329,5 µs
- UDP payload duration: 1500·8/11 µs = 1091 µs

After one frame, an ACK is given:

- Duration of ACK frame (only 1 Mbit/s!): 24·8 µs (PHY H) + 14·8 µs = 304 µs

Now: what happens between data frame and its ACK, and what happens after an ACK?



How long is the backoff? The contention window has 31 slots of 20 µs each. Rough estimation: over time, we get a mean backoff time (MBT), which is 31/2·20 µs = 310 µs.

Transmission sequence: Data frame – SIFS – ACK – DIFS – MBT

→ 1329,5 + 10 + 304 + 50 + 310 = 2003,5 µs

Maximum throughput means: we have successful transmission in each frame, no collisions occur. Then in 2003,5 µs, we can transmit one UDP payload, i.e. we use 1091 µs for data transfer

→ 1091/2003,5 · 11 Mbit/s = **5.99 MBit/s**

Only about 6 MBit/s of the data rate you can make use of for your payload – given that no transmission errors occur and only a single station sends data! This is rarely more than 50%.

What about newer 802.11 versions? Data rates are higher, but the general access schema is the same, leading to similar results. BUT: by introduction of TXOP, a station can transmit several frames by only getting medium access once. Depending on the configured TXOP value, the efficiency can be drastically increased. Frame compression furthermore can reduce the general header overhead. And when changing to OFDMA, medium access is even more efficient. Thus, 802.11 MAC is a good example for protocol evolution.

Exercise 4.2: WLAN Medium Access

Have a closer look at some aspects of the medium access in 802.11.

a.) A station that successfully transmitted a frame and immediately wants to transmit a second frame, does have to choose a random backoff in the next transmission attempt. What is the reason for doing so?

If a station *A* would be allowed to start a second frame only DIFS after the first one, it would get exclusive medium access for some time: probably all other stations would be blocked because they listen to *A*'s transmissions and always have to choose a random backoff to avoid collisions. If station *A* would behave differently, it would be advantaged.

As seen in exercise 4.1, this behaviour in situations of low load makes the medium access very inefficient, but a station does not know about the transmission attempts of other stations, thus it has to be careful.

But also here: by introduction of TXOP, a station in fact can transmit a limited number of frames before having to compete with other stations again.

b.) Each frame in 802.11 has to be acknowledged. In Ethernet, no acknowledgements are used. Why are acknowledgements necessary in 802.11, but not in Ethernet?

You can name at least two reasons. First, in 802.11 we suffer from the hidden station problem. A station has to get an ACK as a confirmation that the receiver had no collision while receiving a frame.

Second, the bit error rate is by far higher. In cabled networks nearly no frames are lost by bit errors, so that error detection and correction on efficiency reasons can be left over to higher layers. But in wireless networks, it is more efficient to immediately deal with error situations on the MAC layer.

c.) *Assume that an 802.11b station is configured to always use the RTS/CTS handshake before transmitting a frame. This station now wants to transmit a frame of 1000 Byte size. How long does it take for the station to successfully transmit the frame?*

Assume that no other station currently is transmitting or wants to transmit, maximum data rate can be maintained, and no bit errors occur. Additionally, neglect the latency on the wireless link.

As in 3.2, have a look at the whole process:

Transmission wish – DIFS – RTS – SIFS – CTS – SIFS – DATA – SIFS – ACK

In 802.11, SIFS is 10µs and DIFS is 50 µs.

From slide III-73, read out frame sizes: 20 byte for RTS, 14 byte for CTS, and 14 byte for ACK. Together with the 1000 byte payload, 1048 byte have to be transmitted – the data frame with 11 Mbit/s, the RTS, CTS and ACK with only 1 Mbit/s. This takes 1111,27 µs.

Also consider that for each frame, using long frame format, 192 µs for the PHY frame header are needed – for our four frames this takes 768 µs.

In sum: 768 + 1111 + 50 + 3*10 = 1,96 ms.

And without RTS/CTS? Only consider DIFS – DATA – SIFS – ACK, ending up with

50 + 192 + 727 + 10 + 192 + 112 = 1,28 ms.

So you can see: if collisions are not very probable or only short frames are sent, using the RTS/CTS handshake is very inefficient. On this reason by default the RTS/CTS mode is often disabled.

d.) 802.11 allows for mobility support – if a station moves between several BSSs of the same ESS, the distribution system always keeps track of where a station currently resides.

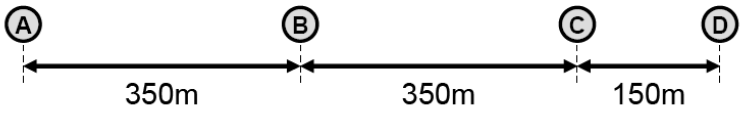
Assume that all APs of an ESS are connected with a single switch as distribution system. Sometimes, an AP has to transmit a frame with a faked MAC address to allow the switch to correctly forward the following frames. Why is this necessary?

The switch has its own forwarding table. It has no mobility management integrated but just learns which MAC addresses are placed behind which ports.

If a station is mobile and joins a new AP, the switch's entry for the MAC address of this station becomes invalid, but only would be updated on the next transmission attempt of this station. If data has to be delivered to this station, it in the meantime still would be sent to the old AP. On this reason, the new AP has to broadcast a frame with the station's MAC address as source address, so that the switch immediately adjusts its entry.

Exercise 4.3: WLAN in Ad-hoc Mode

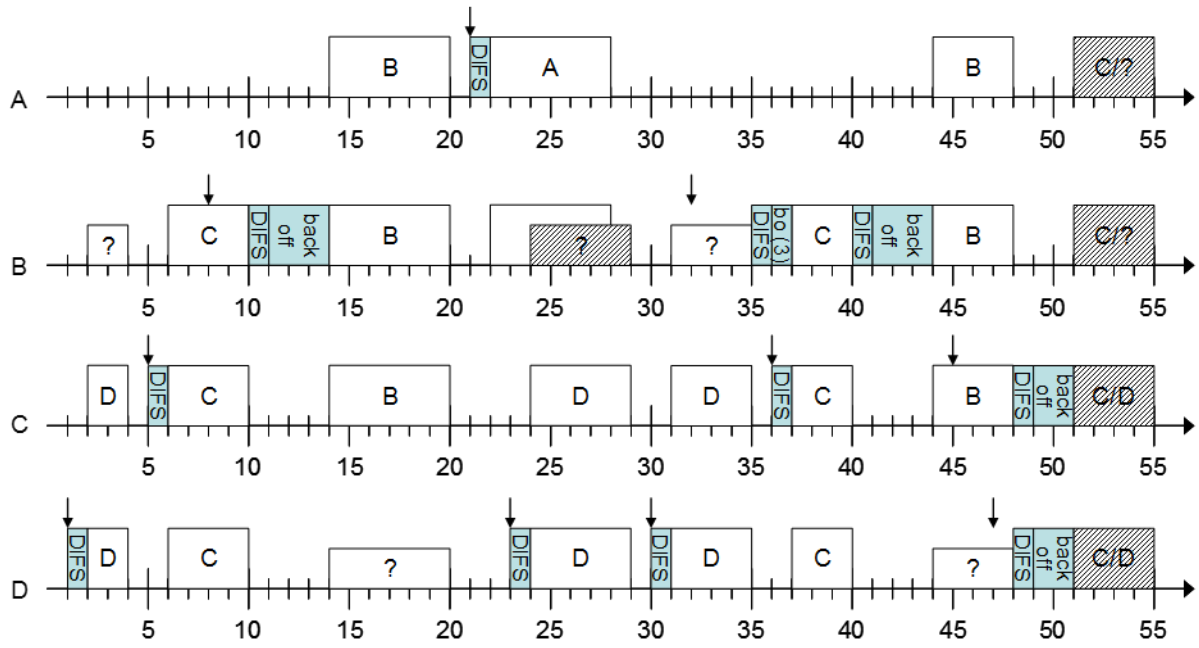
Given is the following ad-hoc network based on WLAN:



The transmission range of all stations is 375m, the interference range is 670m. The stations are sharing the same channel and have sending wishes as shown in the following table:

Sender	Receiver	Time	Sending duration	Chosen backoff
D	**C**	**1**	**2**	**-**
C	**D**	**5**	**4**	**-**
B	**A**	**8**	**6**	**3**
A	**B**	**21**	**6**	**-**
D	**C**	**23**	**5**	**-**
D	**C**	**30**	**4**	**-**
B	**A**	**32**	**4**	**4**
C	**D**	**36**	**3**	**-**
C	**Broadcast**	**45**	**4**	**2**
D	**C**	**46**	**3**	**2**

Assume that the DIFS interval has a duration of 1 time unit. In the following time diagram, sketch the resulting medium use, assuming that unreliable data transfer is used, i.e. no ACKs are sent, no retransmission is done in case of an error.



That shows: at time 24, station D can successfully send to station C, but because of interference range, the transmission of A to B is destroyed. Also, at time 51 you can see that even the backoff does not protect against collision. And worst: at time 37, station B has to stop the backoff counter, because a station which started listening later, but fortunately has not to use a backoff, blocks station B – so the mechanism is not really fair.

Assignment 5

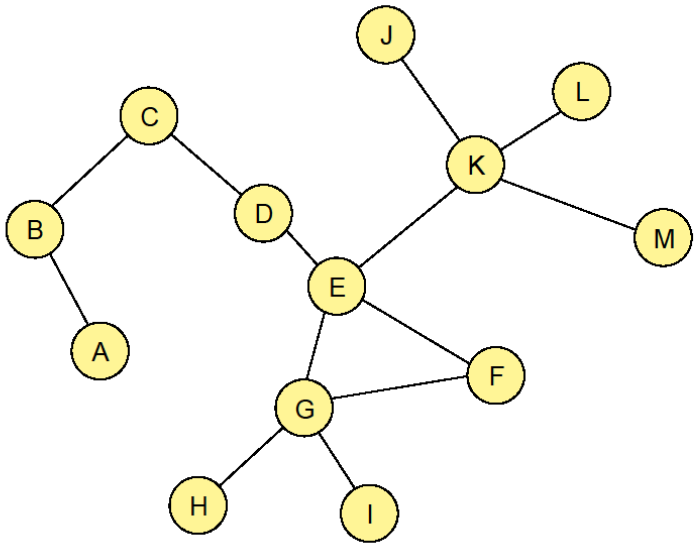
Discussion: 24.06.2026

Exercise 5.1: Routing in wireless mesh networks

a.) *Think of ad-hoc networks with fast moving nodes, e.g., cars in a city. What problems arise even for the routing algorithms adapted to ad-hoc networks? How is the situation changed on highways?*

When cars inside a city move fast, the topology changes are too fast for all routing algorithms to adapt to; cars can enter or leave a street at each crossing, which is quite often. Flooding could be the only solution. On a highway the situation is astonishingly much better, even if speed is higher: you can think about groups of cars with similar speed and same direction. Forwarding is possible along the roads in driving direction.

b.) *Given is a mesh network which initially has the following topology:*



Nodes connected in this picture are in transmission range and can forward messages along the edges. For routing, DSR should be used. The caches of all nodes are empty. Now, D wants to send some packets to H. Give the sequence of messages exchanged for finding a path. For each message, give sender, receiver, and the list of passed nodes as written in the header by DSR. Assume that the paths are symmetrically.

Basically, the messages transmitted are containing more information than demanded here: (address of requesting node, address of searched node, unique ID, list of traversed nodes). This means, D would start path discovery by broadcasting “(D, H, 4242, D)”. Because the first three entries are the same for all forwarded messages (and allow each node to decide if it is the destination and if it has received the message on another path before), we are skipping them here. We only write down list list of traversed nodes, to records the path:

- *D* starts searching by broadcasting a message to all its neighbors: *D → C, E: (D)*
- *C* and *E* read out that they are not the designated receiver, and go on broadcasting:
  - *C → B, D: (DC)*
  - *E → D, F, G, K: (DE)*
- *B*, *F*, *G*, *K* forward (D detects the message as duplicate and discards it):
  - *B → A, C: (DCB)*
  - *F → E, G: (DEF)*
  - *G → E, F, H, I: (DEG)*
  - *K → E, J, L, M: (DEK)*
- *A*, *I*, *J*, *L*, *M* are broadcasting on (C, E, F, and G detect the message as duplicate and discard)
  - *A → B: (DCBA)*
  - *I → G: (DEGI)*
  - *J → K: (DEKJ)*
  - *L → K: (DEKL)*
  - *M → K: (DEKM)*

  Note that all these nodes reach no new nodes any more, but maybe the nodes do not know about possible connections to other nodes before a discovery starts, so they all do just not know that broadcasting makes no sense.
- *H* sends a route reply to *D* along the path (because we assume it to be symmetrically):
  - *H → G: (G, E, D – DEGH)*
  - *G → E: (E, D – DEGH)*
  - *E → D: (D – DEGH)*

  Note that the information in brackets now shold have the meaning: (source routing information to bring the data to D – payload, that is: discovered path as information for D).

c.) *What happens if the connection between E and G breaks?*

Breakdown of the connection between *E* and *G*: *E* generates a route error when trying to transmit *D*'s next message to *H*, thus *D* starts all over with flooding a route request. Or, depending on configuration: *E* starts route discovery and sends the update back to *D*.

Exercise 5.2: GSM: Cells and Channels

The moon recently has been populated, but till now nobody has installed a telephony system. Thus you decide to install the first GSM system on the moon. You have given the following premises:

- *The given frequency band is 880.0 – 882.8 MHz and 935.0 – 937.8 MHz.*
- *You use a cluster size of 7.*
- *Only full traffic channels are used.*
- *Only one TDMA channel per cell is needed for signalling and control.*
- *There are 12.3 million inhabitants, with constant population density in the whole coverage area.*
- *Each inhabitant will use the telephony service with an average probability of 1%.*

Give an estimation: how many cells are needed in your installation?

You have given a bandwidth of 2.8 MHz. Because one channel has to have a bandwidth of 200 kHz, you have available at all **14 frequency channels**.

Per cell in the cluster, you can thus spend **2 frequency channels**.

Per frequency channel you have 8 TDMA channels, thus you could install 16 TCHs per cell. But, consider control overhead: at most, you can use 15 TCHs and have to spend the rest for control information. (Very simplified, but eases calculations.) Thus, there are **15 simultaneous calls per cell** possible.

If every inhabitant does a call with a probability of 1%, you have always to expect 123.000 simultaneous calls. Simple division: 123.000 divided by 15 calls per cell gives the necessity for 8.200 cells, i.e. (assuming one BTS per cell) **8.200 BTSs**.

Exercise 5.3: GSM System

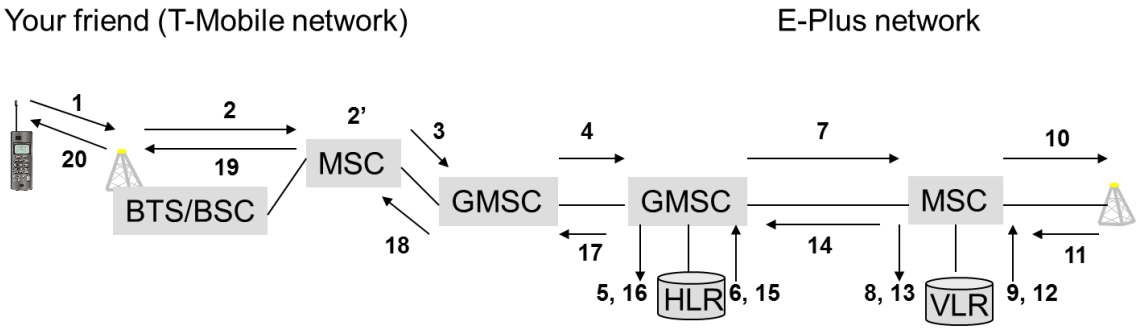
a.) *Given is the following situation: you are subscribed to the GSM network of E-Plus. You are travelling to Kraków; on arrival, you switch on your mobile phone and log in into the Orange GSM network. Nearly at the same time, a friend who is subscribed with T-Mobile, tries to call you. What happens in the following two situations:*

(i) the friend tries to call you before you switched on your phone.

(ii) the friend tries to call you after you switched on your mobile phone.

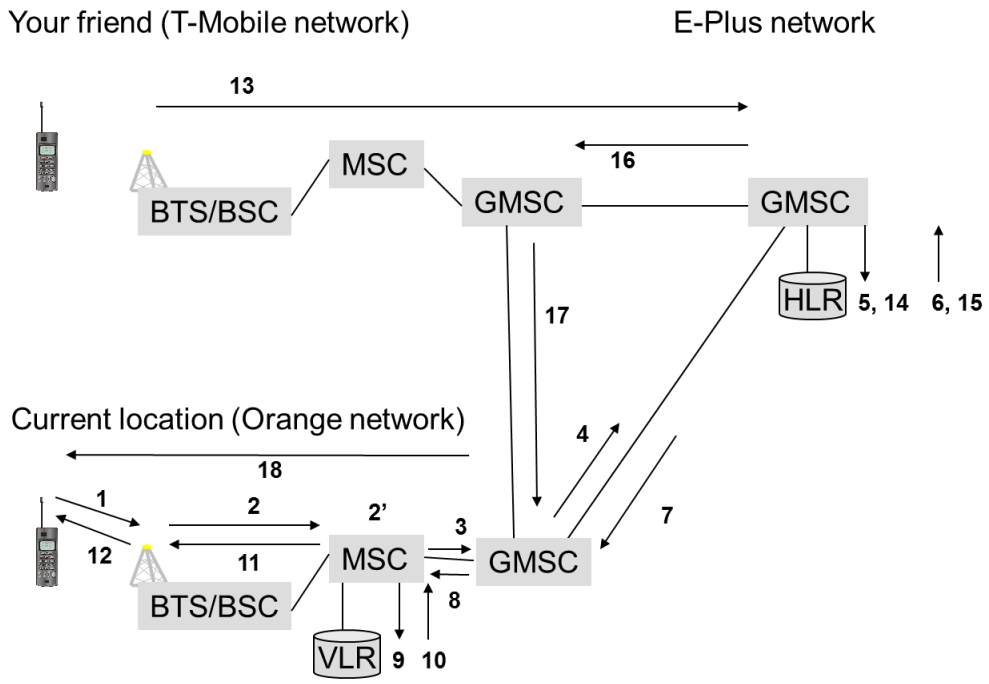
Sketch both scenarios with all involved GSM subsystems.

First scenario:



- 1,2: request of switching a call from the t-mobile customer to its MSC (via BTS/BSC)
- 2': the MSC would locally check if it knows the called number, i.e. if this number currently is registered at the MSC. In our case it is not. Note that a central registry has to be consulted to find out to which network the number belongs. This mechanism is not included here because it is country-specific. Just assume that we know the destination network.
- 3: route request to GMSC to hand it over to the correct network.
- 4: forwarding of the request to the home network of the callee.
- 5, 6: HLR is asked for the current location. Now there are two options:
  - The network frequently checks if a customer still is within the location area stored in HLR and the corresponding VLR: the responsible MSC initiates a paging to an inactive station, and if no response is given, the location information is set to "unknown". This information is propagated to the HLR. Probably this already has happened. Continue with 17.
  - A location area is still stored in HLR and corresponding VLR. This is unlikely after some hours with switched-off phone, but more interesting. ☺ Continue with 7.
- 7 – 9: forwarding of the request to the last responsible MSC, reading out a location area identifier of the callee from VLR
- 10, 11: paging of the callee in all cells of the location area. It comes back no answer
- 12, 13: deletion of the callee's data from the VLR
- 14 – 16: inform the HLR about the absence. HLR is changing location information to n/a
- 17 – 20: give back absence information to the caller. No call established.

Second scenario:



Now start with own phone:

- 1, 2: set up channel to next MSC. Now this MSC has to look up to which network your phone belongs (2')
- 3, 4: forward request to join to home network of our mobile phone, together with our current location information
- 5 – 8: our HLR first checks permissions: are we allowed to log in into Orange. We assume that we are. The HLR now stores the location area which was told by the request. It grants the asking MSC the permission to let join the device and to create billing information.
- 9, 10: foreign MSC stores our information in its VLR
- 11, 12: the device is granted access, it now has joined the network.
- 13: our fried tries to establish a call. This corresponds to the above steps 1 – 4.
- 14 – 16: our HLR gives a link to a foreign MSC.
- 17: T-mobile's GMSC switches the call to Orange's GMSC
- 18: The call is delivered as in steps 7 - 10 in the above scenario. But this time, we can answer the paging and pass back a positive response all the way to our friend. Our phone starts ringing.

b.) *For data transmission, on average a higher transmit power is used than for voice transmission. Why?*

In voice transmission (even with compressed data) more errors can be tolerated than in data communication. To reduce the error risk in data transmission, the transmit power is increased to improve the SNR.

c.) *Consider a handover situation. What problems from a user's point of view could occur with a handover if you have an active HSCSD or GPRS connection?*

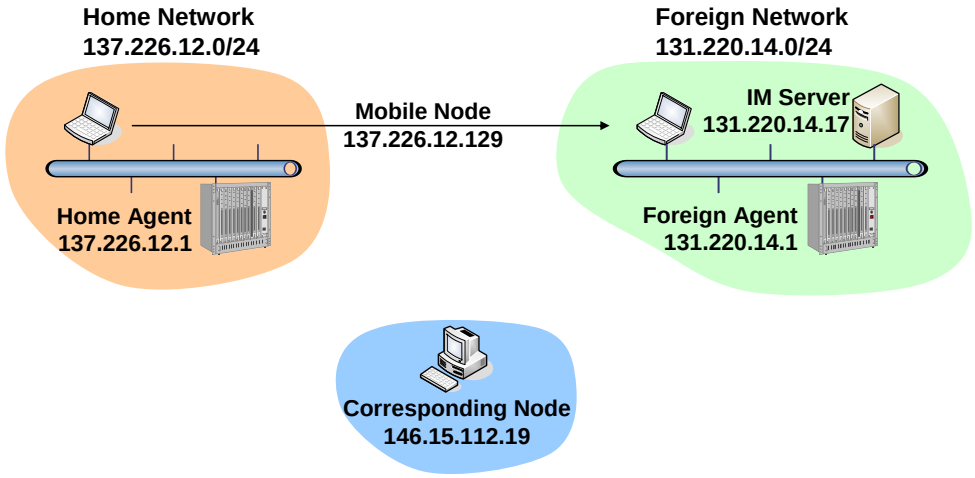
After handover there is no guarantee for the same data rate as in the old cell; maybe the new cell is overloaded.

Assignment 6

Discussion: 08.07.2026

Solution 6.1: Mobile IP Scenario

a.) Have a look at the following scenario:



User A at host 146.15.112.19 wants to send a message to user B at host 137.226.12.129. For delivery of the message it should be used an instant message server with address 131.220.14.17. User B is mobile and currently wants to join the foreign network 131.220.14.0/24.

How is data in general delivered from user A to user B? Give all actions performed by the involved components, beginning with the mobile node joining the foreign network, and additionally describe the message flow for delivering data to B. Distinguish between the two modes supported by Mobile IP (FA-managed CoA and co-located CoA).

What is the flow of an instant message from user A to user B if the foreign agent manages the CoA?

Joining the network, registering with home network:

- 137.226.12.129 (named *B*) comes into network range of 131.220.14.0/24 and listens for agent advertisements. It receives an agent advertisement from 131.220.14.1 (named *FA*); this advertisement carries a list of CoA addresses.
- *B* chooses a *CoA* (e.g. 131.220.14.9, but let us just call it CoA)
- *B* registers via FA with 137.226.12.1 (named *HA*). During registration, *B* is authenticated by *HA*. If authentication is successful (in our case it is), *B* is has joined the foreign network. It is listed within the FA, and in case of a co-located CoA, the FA removes the chosen CoA from its pool of addresses.

Delivering data to *B*:

- All data from 146.15.112.19 (named *A*) or any other communication partner is just routed to 137.226.12.0/24.
- *HA* intercepts all packets for *B* and tunnels them to *CoA*.
- In case of co-located *CoA*: the tunnel ends at the mobile device itself. The data is delivered to our mobile host, but on *CoA*. It checks the protocol field of the received packet and finds IP-in-IP. It decapsulates a new IP packet and forwards it to its destination address, which is *B*. (Local forwarding of data.)
- In case of a foreign-agent-managed CoA, *FA* is end of the tunnel and decapsulates the original packet It has a look to the destination address and finds *B* – which it knows to be local because of the former registration.

Using instant messaging:

- *A* sends message to IM server (131.220.14.17).
- IM server sends data on to *137.226.12.129*. It does not know that this host is locally connected, but just sends the packet to the entrance router of 131.220.14.0/24.
- The entrance router MIGHT be the *FA*. In case it is the same, the entrance router knows *B* and can directly forward the data. In case it is not the same, the *FA* does not get the packet, but the entrance router routes it to 137.226.12.0/24. Here *HA* intercepts it and tunnels it to *CoA* – which is on host *FA*. It is then locally forwarded to *B*. We have most extreme triangular routing.

b.) *Assume that a mobile node joins a foreign network which uses NAT. Could there be any problems with Mobile IP in presence of NATs? If yes, which problems do arise?*

For sure there will be problems. First, the MN itself tells the home agent its CoA in its registration request. The CoA is a private address, thus the HA discards the registration request.

Second, the tunnel between HA and CoA would have to cross the NAT router. But, using e.g. IP-in-IP encapsulation, the NAT router will find no TCP/UDP header for its translations.

For both problems, there are specified extensions, but they weren't treated in the lecture.

c.) *What are main advantages of Mobile IPv6 compared to Mobile IPv4?*

One advantage is that there is no FA any more so that any network which lets you join can be used as foreign network; no infrastructure support is needed.

But, more important: mobility is integrated as IP options. Route optimization is possible by using "location" IP addresses as source- and destination in the IP header, and by including the "identity" IP address as an option which is applied before the packet is processed on receiver side.

And also security is integrated.

Solution 6.2: Mobile IP, HIP, and xG networks

a.) *Compare the main components of Mobile IP with xG networks – is there any concept in xG networks corresponding to Home Agent, Foreign Agent, host's original address, and Care-of-Address? What is similar to these concepts, where are differences? Does xG networks offer functionality not implemented in Mobile IP or vice versa?*

Used here: GSM terminology for xG networks:

- Home Agent: corresponds to HLR / GMSC. Similarity: entry point to the network, stores current location of all mobile nodes belonging to the own network, forwards data. Differences: Much larger network (more users) managed by HLR; HA only is responsible for small network. Consequently, in xG mapping of addresses to locations and forwarding of data are separated because of scalability. And: HLR only stores location area, while HA stores concrete address (which is a location area identifier only if it is managed by the FA).
- Foreign agent: corresponds to MSC with VLR. Similarity: authenticate and manage all locally connected mobile users. Difference: FA only manages foreign devices. In VLR, all connected devices are managed, also the ones of this home network.
- Original address: corresponds to phone number. Similarity: static ID of device/user. Difference: misuse of home locator as ID in Mobile IP. In xG, concepts of locator and ID are clearly separated.
- CoA: corresponds to Location Area ID / Cell ID. Similarity: gives location information. Difference: especially in case of co-located CoA it identifies a certain device, not a location area.

Given in xG networks but missing in Mobile IP: integrated security! Mobile IP needs IPSec for encryption and also some trust schema for authentication. And furthermore there is lots of more functionality in xG networks just because it was designed for mobility: handover procedures with very low latency, additional components like a maintenance center, …

b.) *Can you find the indirection point of Mobile IP (i.e. the home agent) in the HIP architecture? If yes, where is it? If no, why not?*

The home agent manages the current location of a mobile node and redirects all packets.

In terms of HIP, managing the location information corresponds to the storage of a mapping of a HI to an IP address. We can use DNS or rendezvous servers or any other registry to resolve this mapping. In this regard, there is still some infrastructure support necessary when using HIP.

But, there is no separated indirection point for data in HIP. The registry only resolves the mapping but does not do data forwarding. The indirection nevertheless takes place, but by a local replacement of addresses on the receiving device.

Thus in HIP we have a separation between mapping of addresses/locations and forwarding of data – in this regard. HIP is more comparable with xG (HLR to map addresses only) than with Mobile IP (data indirection).